Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Kinematics

MARKSCHEME ($g = 9.8\,\mathrm{m\,s^{-2}}$)

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1. $v = u + at = 10$ **M1**; $s = ut + \frac{1}{2}at^2 = 25$ **A1** [4]
2. $a = \frac{0-20}{4} = -5$ **M1**; $s = \frac{20+0}{2} \times 4 = 40$ **A1** [4]
3. $v^2 = u^2 + 2as = 16 + 60 = 76 \Rightarrow v = \sqrt{76}$ **M1A1** [4]
4. $0 = u^2 - 2gh \Rightarrow h = \frac{225}{19.6}$ **M1A1** [4]
5. $s = \frac{1}{2} \times 1.5 \times 36 = 27$ **M1A1** [3]
6. $t = \frac{30}{2} = 15$ **M1**; $s = \frac{30}{2} \times 15 = 225$ **A1** [4]
7. $v = 5 - 6 = -1$ **M1**; $s = 5 \times 3 - \frac{1}{2} \times 2 \times 9 = 6$ **A1** [4]
8. $s = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{90}{9.8}}$ **M1A1** [3]
9. $a = \frac{15}{5} = 3$ **M1**; $s = \frac{10+25}{2} \times 5 = 87.5$ **A1** [4]
10. $100 = \frac{1}{2}a \times 25 \Rightarrow a = 8$; $v = 20$ **M1A1** [4]
11. $t = \frac{2u}{g} = \frac{40}{9.8}$ **M1A1** [3]
12. $v^2 = u^2 + 2as \Rightarrow a = 1$; $t = \frac{v-u}{a} = 6$ **M1A1** [5]
13. $v^2 = u^2 + 2as \Rightarrow a = -3$ **M1A1** [3]
14. $v = u + at$; $s = ut + \frac{1}{2}at^2$ **B1B1** [3]
15. $v^2 = 144 + 2 \times 9.8 \times 16$ **M1**; $v \approx 18.0$ down **A1** [5]
16. $20 = 2u + 2a$; $50 = 3(u + 2a) + \frac{1}{2}a \times 9$ (or equiv.) **M1M1**; solve $u = 5, a = 5$ **A1** [6]
17. Accel 100; const 400; decel 200; total 700 **M1A1A1** [6]
18. $s = -30 = 25t - \frac{1}{2} \times 9.8t^2$ **M1**; solve quadratic **M1**; $t \approx 6.05$, $v \approx -34.3$ **A1** [6]
19. $10t = t^2 \Rightarrow t = 10$, $s = 100$ **M1A1** [6]
20. $a = 3$; 4th second $= u_3 + \frac{1}{2}a = 13.5$ **M1A1** [5]
21. $s = 600 + 225 = 825$ **M1A1** [4]
22. $T = \frac{2u}{g}$; $H = \frac{u^2}{2g}$ **M1A1** [5]
23. Accel $s = 24$; $v = 12$; decel $t = 6$, $s = 36$; totals 10 s, 60 m **M1A1** [6]
24. $u = 3, a = 4$; $v = 23$ **M1A1** [4]
25. $-h = 5 \times 4 - \frac{1}{2} \times 9.8 \times 16$ **M1**; $h = 58.4$ **A1** [6]